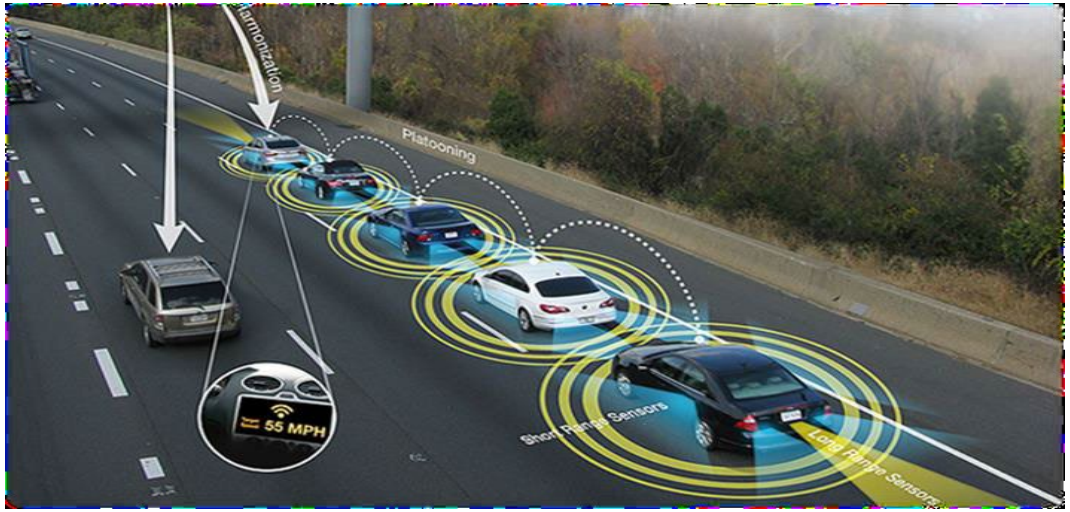
Lingxi Li
Professor of Electrical and Computer Engineering
Indiana University–Purdue University Indianapolis

Distributed stochastic model predictive control for heterogeneous vehicle platooning under uncertainty

Vehicle platooning for connected and automated vehicles (CAVs) has many potential benefits, such as lowering fuel consumption, improving traffic safety, and reducing traffic congestion. However, challenges remain toward safe and efficient vehicle platooning since its performance could be degraded due to uncertainty from vehicle dynamics, environmental disturbances, and communication delays, among others. In this talk, I will introduce one recent work on a new control method that combines distributed stochastic model predictive control (DSMPC) with Taguchi's robustness (TR-DSMPC) for vehicle platooning. The proposed method inherits the advantages of both Taguchi's robustness (maximizing the mean performance and minimizing the performance variation due to uncertainty) and stochastic model predictive control (ensuring a specific reliability level). The proposed method was compared with two other MPC-based methods in terms of safety (spacing error) and efficiency (relative velocity). The results indicate that the proposed method can effectively reduce the performance variation and maintain the mean performance compared to other methods.

icensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restric

IEEE ITSS Workshop on Connected and Autonomous Vehicles



Distributed Stochastic Model Predictive Control for Heterogeneous Vehicle Platooning under Uncertainty

Lingxi Li
Transportation and Autonomous System Institute (TASI)
Department of Electrical and Computer Engineering
Purdue School of Engineering and Technology, IUPUI, USA
October 27, 2022

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Presentation Outline

1. Introduction and Background
2. Platoon Modeling
3. Control Methods
4. Simulation Results
5. Conclusions and Future Work

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Introduction and Background (1)

- **Vehicle Platooning**
  - **Basic Concept**
    - Vehicle platooning mainly considers the scenario where several vehicles drive in the same lane
      - Adjust the inter-vehicle distance
      - Maintain the desired speed by both longitudinal and lateral control techniques
  - **Potential Benefits**
    - Improve traffic efficiency
    - Enhance road safety
    - Reduce overall fuel consumption

USA-PATH

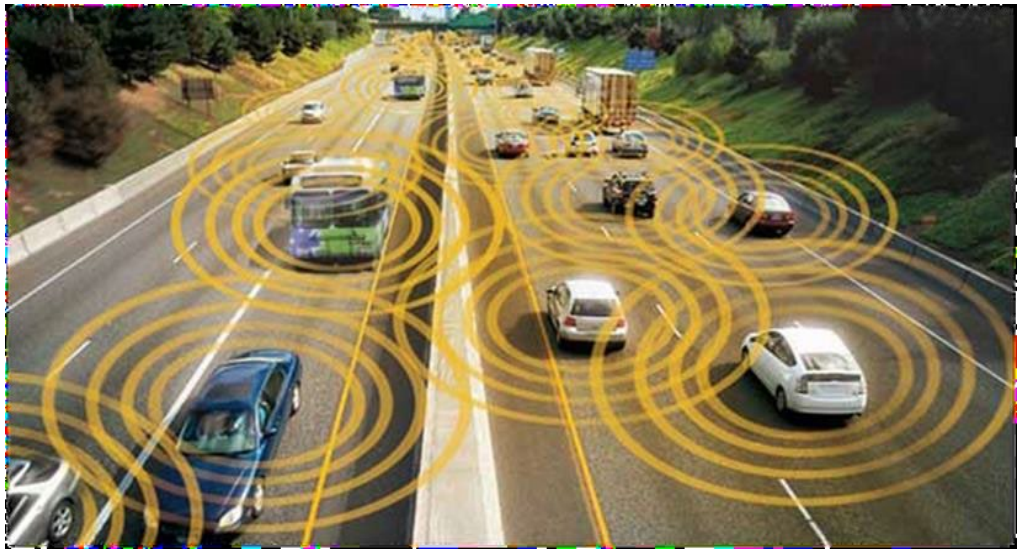
Europe -SARTRE

Japan -Energy ITS

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Introduction and Background (2)

- **Vehicle Platooning Characteristics**
  - **Homogeneous vs. Heterogeneous**
    - Homogeneous if all vehicles have identical dynamics
    - Heterogeneous if vehicles have non-identical dynamics
  - **Typical Communication Topology**
    - Predecessor-following (PF)
    - Predecessor-leader following (PLF)
    - Bidirectional (BD)
  - **Typical Control Methods**
    - Model Predictive Control (MPC)
    - H-Infinity Control
    - Min-max robust MPC
  - **Uncertainty**
    - From the vehicle itself and/or the external environment
    - Undermine nominal-based control method



Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Introduction and Background (3)

- **Research Question**
  - How to deal with uncertainty for vehicle platooning control?
    - Uncertainty might come from varying road conditions, change of vehicle dynamics, or disturbance from the driving environment. Modeled by stochastic processes.
- **Taguchi's Robustness**
  - **Basic Idea**
    - Minimize the effects of uncertainty in the design objective without eliminating the source of uncertainty
  - **Quality Loss**
    - Maximize the insensitivity of the response to uncertainty and their average performance
    - First-Order Second Moment (FOSM) is used for uncertainty quantification of vehicle dynamics instead of sampling-based methods to reduce the computational cost
  - **Advantages**
    - Maximize the mean performance
    - Minimize the performance variations due to uncertainty

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Introduction and Background (4)

- **Main Contributions**
  - **Novel Control Strategy**
    - Taguchi's robustness – Ensuring robustness
    - Distributed stochastic model predictive control – Ensuring reliability
  - **New Uncertainty Propagation**
    - First-order Second Moment was used
    - More efficient than Monte Carlo-based sampling methods

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Problem Formulation (1)

- **Model of Vehicle Dynamics**
  - **Assumptions**
    - Considering vehicle longitudinal dynamics only
    - Focusing on the flat road condition and normal driving states
  - **Discrete-time (D-T) dynamic model**
    - D-T dynamic model of any node in the following vehicle

$$\begin{cases} s_i(k+1) = s_i(k) + v_i(k)\Delta t \\ v_i(k+1) = v_i(k) + \dfrac{\Delta t}{m_i}\left(\dfrac{\eta_i T_i(k)}{r_i} - C_{a_i} v_i(k)^2 - m_i g f_i\right) \\ T_i(k+1) = T_i(k) + \dfrac{\Delta t}{\tau_i}(u_i(k) - T_i(k)) \end{cases}$$

$$x_i(k+1) = A_i x_i(k) + B_i u_i(k) + \psi_i$$

$$x_i = (s_i, v_i, T_i)^{\mathrm{T}}$$

$$A_i = \begin{bmatrix} 1 & \Delta t & 0 \\ 0 & 1 - \dfrac{C_{a_i} v_i(k)\Delta t}{m_i} & \dfrac{\eta_i \Delta t}{m_i r_i} \\ 0 & 0 & 1 - \dfrac{\Delta t}{\tau_i} \end{bmatrix}$$

$$B_i = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\tau_i} \end{bmatrix}$$

$$\psi_i = \begin{bmatrix} 0 \\ -g f_i \Delta t \\ 0 \end{bmatrix}$$

- i represents all following vehicles
- $s_i(k)$: displacement of vehicle i at time k
- $v_i(k)$: velocity of vehicle i at time k
- $T_i(k)$: coupled torque of driving & braking of vehicle i at time k

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Problem Formulation (2)

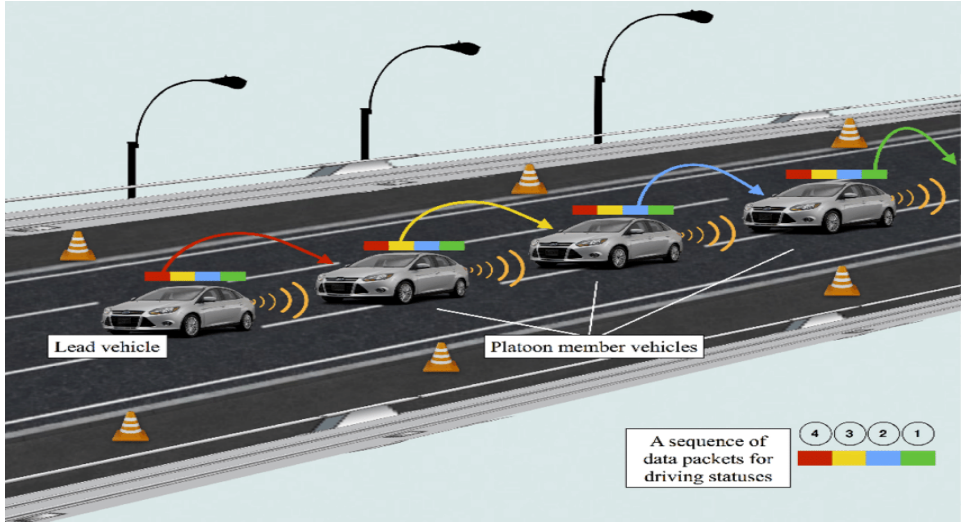
- **Platoon Model**
  - **Overall discrete-time model of vehicle platoon for N following vehicles**

$$X(k+1) = AX(k) + BU(k) + \Psi$$

where:

$$\begin{cases} X(k) = [x_1^T(k), x_2^T(k), \cdots, x_N^T(k)]^T \\ Y(k) = [y_1^T(k), y_2^T(k), \cdots, y_N^T(k)]^T \\ U(k) = [u_1^T(k), u_2^T(k), \cdots, u_N^T(k)]^T \end{cases}, i \in N$$

where $\boldsymbol{X} = (\boldsymbol{x}_1; \ldots; \boldsymbol{x}_N) \in \mathbb{R}^{3N\times 1}$, $\boldsymbol{U} = (\boldsymbol{u}_1, \ldots, \boldsymbol{u}_N)^{\mathbf{T}}$, $\boldsymbol{A} = diag(A_1, \ldots, A_N) \in \mathbb{R}^{3N\times 3\ N}$, $\boldsymbol{B} = diag(B_1, \ldots, B_N) \in \mathbb{R}^{3N\times N}$, and $\boldsymbol{\Psi} = (\psi_1; \ldots; \psi_N) \in \mathbb{R}^{3N\times N}$.



- X(k): vector of vehicle state variables at time k
- Y(k): vector of vehicle outputs at time k
- U(k): vector of vehicle control inputs at time k

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Problem Formulation (3)

- **First Order Second Moment**
  - **Definition**
    - An approximation method to propagate the uncertainty from input space to output space
    - Based on the performance function simplified by the 1st-order Taylor expansion
  - **Formula**
    - Performance function

$$Y = g(Z)$$

where Z is a vector to represent input random variables, and Y is the model response

- Linearization

$$Y = g(Z) \approx g(\mu_Z) + \nabla g(\mu_Z) \cdot (Z - \mu_Z)$$

- Variance of the response

$$\sigma_Y^2 = \sum_{l=1}^{n} \left(\frac{\partial g}{\partial Z_l} \right)^2 \Bigg|_{\mu_Z} \sigma_{Z_l}^2$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Problem Formulation (4)

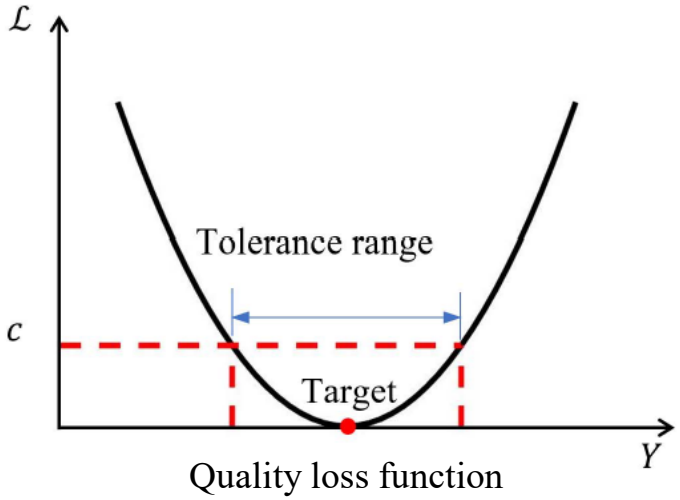
- **Taguchi's Robustness**
  - **Taguchi's quality loss**
    - Nominal-the-best type performance

$$\mathcal{L}(Y) = \sum_{i=1}^{m} w_i \left(Y_i - T_{ri}\right)^2$$

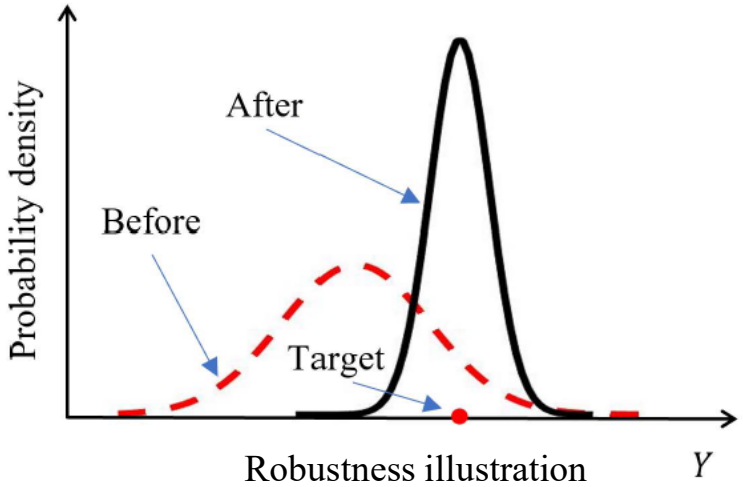
where T_r is the design targets, w_i is the weighting factor determined by cost

  - **Modified quality loss**
    - Adding variance of system response

$$E_{\mathcal{L}(Y)} = \sum_{i=1}^{m} w_i \left[\left(\mu_{Y_i} - T_{ri}\right)^2 + \sigma_{Y_i}^2\right]$$



Quality loss function



Robustness illustration

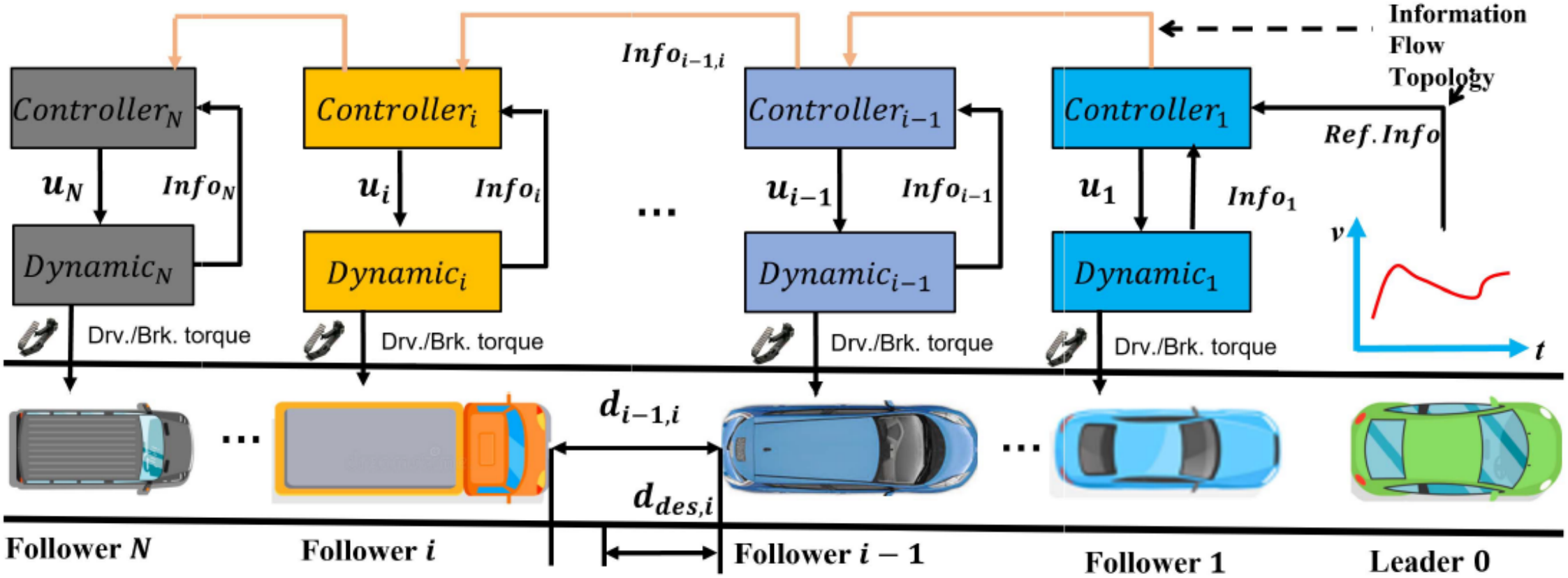
where $E_{\mathcal{L}(Y)}$ is the expected quality loss

- Ensure the mean target performance and reduce the performance variations

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Control Methods (1)

- **Design of Distributed SMPC with Taguchi's Robustness (TR-DSMPC)**
  - **Overall Control Structure**



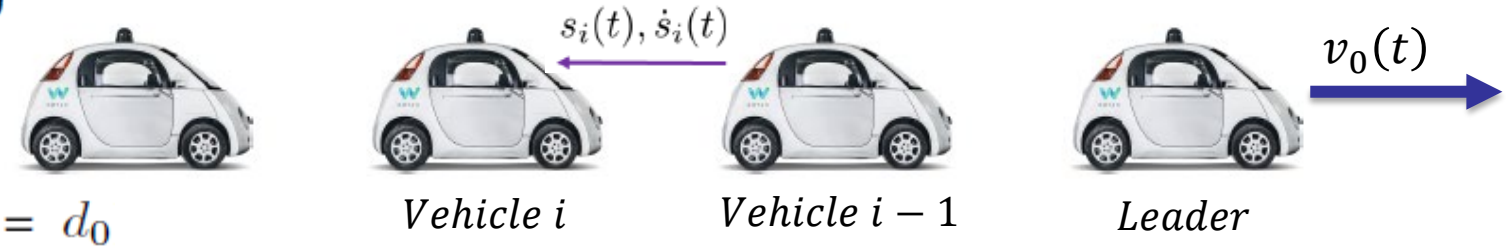
Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Control Methods (2)

- **Design of Distributed SMPC with Taguchi's Robustness (TR-DSMPC)**
  - **Control Objectives**
    - Follow the lead vehicle's velocity while tracking
    - Maintain the desired gap between any two consecutive vehicles

$$\begin{cases} \lim_{t\to\infty} \|v_i(t) - v_0(t)\| = 0 \\ \lim_{t\to\infty} \|s_{i-1}(t) - s_i(t) - d_{i-1,i}\| = 0 \end{cases}, \quad i \in \mathcal{N}$$

$v_0(t)$: Leader's speed



Constant Spacing Policy: $d_{i-1,i} = d_0$

    - Uncertainty considered
      - Vehicles: Mechanical efficiency coefficient and inertial time lag
      - Outside environment: aerodynamic drag coefficient and rolling friction coefficient

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Control Methods (3)

- **Design of Distributed SMPC with Taguchi's Robustness (TR-DSMPC)**
  - **Uncertain Modeling**
    - Time-dependent uncertainty
      - Modeled by stochastic processes
      - Values of the process at two-time instants are correlated with respect to the distance or time
      - Simulate parameters related to road conditions such as friction factor and air drag coefficient
      - Auto-correlation function (square exponential kernel)

$$\rho_{1,2} = \exp\left\{-\left(\frac{|t_1 - t_2|}{\theta_t}\right)^2\right\}$$

where θ is the correlation length along the time

    - Karhunen-Loeve (K-L) expansion

$$H(t) = \mu(t) + \sum_{i=1}^{M} \sqrt{\lambda_i}\varphi_i(t)\xi_i$$

Assume parameter uncertainties are Gaussian random fields

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Control Methods (4)

- **Design of Distributed SMPC with Taguchi's Robustness (TR-DSMPC)**
  - **Objective Function**
    - Desired state for each follower

$$\begin{cases} s_{d,i}(k) = s_0(k) - i d_0 \\ v_{d,i}(k) = v_0(k) \\ T_{d,i}(k) = C_{a_i} v_0(k)^2 + m_i g f_i \end{cases}$$

    - Safety, efficiency, and control

$$E_{J_i}\left(y_i(k), u_i(k)\right) = \sum_{j=0}^{N_p-1} \left(\Delta y_i^2(k+j+1) + \Delta u_i^2(k+j+1) + \Delta y_{a,i}^2(k+j+1)\right.$$

    - Variations of safety, efficiency, and control

$$V_{J_i}\left(y_i(k), u_i(k)\right) = \sum_{j=0}^{N_p-1} \Big[w_{\Delta y_i} \sigma_{\Delta y_i}^2 (k+j+1) + w_{\Delta y_{a,i}} \sigma_{\Delta y_{a,i}}^2 (k+j+1) + \sigma_{\Delta u_i}^2 (k+j+1) \Big]$$

    - In summary

$$J_i\left(y_i(k), u_i(k)\right) = E_{J_i}\left(y_i(k), u_i(k)\right) + V_{J_i}\left(y_i(k), u_i(k)\right)$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Control Methods (5)

- **Design of Distributed SMPC with Taguchi's Robustness (TR-DSMPC)**
  - **Constraints**
    - Vehicle dynamic constraints and control signal bounds

$$\begin{cases} u_{lb,i}(k+j+1) < u_{p,i}(k+j+1) < u_{ub,i}(k+j+1) \\ \dot{x}_i(j+1) = A_i x_i(j) + B_i u_i(j) + C_i \\ j = 0, 1, \ldots, N_p - 1 \end{cases}$$

    - Probability/reliability constraints

$$\begin{cases} -\delta\sigma_{\Delta s_i} < s_i\left(k+N_p\right) - s_{d,i}\left(k+N_p\right) < \delta\sigma_{\Delta s_i} \\ -\delta\sigma_{\Delta v_i} < v_i\left(k+N_p\right) - v_{d,i}\left(k+N_p\right) < \delta\sigma_{\Delta v_i} \\ -\delta\sigma_{\Delta u_i} < T_i\left(k+N_p\right) - T_{d,i}\left(k+N_p\right) < \delta\sigma_{\Delta u_i} \end{cases}$$

Based on the standard deviation of the spacing error, relative velocity, and control error. A scaling factor δ is also considered, which means that the above three terms are within the 2δ standard deviation of the desired state

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Control Methods (6)

- **Design of Distributed SMPC with Taguchi's Robustness (TR-DSMPC)**
  - **Control Optimization Formulation**
    - Local optimization for platoon control

Local Optimization Problem

$$J_i\left(y_i(k), u_i(k)\right) = E_{J_i}\left(y_i(k), u_i(k)\right) + V_{J_i}\left(y_i(k), u_i(k)\right)$$

→ **Objective Function**

$$u_{lb,i}(k+j+1) < u_{p,i}(k+j+1) < u_{ub,i}(k+j+1)$$

→ **Control input constraint**

$$\dot{x}_i(j+1) = A_i x_i(j) + B_i u_i(j) + C_i$$

→ **System dynamic constraints**

$$y_i^p(N_p|t) = \frac{1}{\Gamma}\sum_{j=\Gamma}\left(y_j^a(N_p|t) - d_{des,ij}\right)$$
$$T_i^p(N_p|t) = h_i(N_p|t)$$

→ **Terminal Constraints**

Constant speed in the end

System final output is around the equilibrium

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Control Methods (7)

- **Design of Distributed MPC**
  - **Robust Design with Reliability Constraints**

Local Optimization Problem

$$J_i\left(y_i(k), u_i(k)\right) = E_{J_i}\left(y_i(k), u_i(k)\right) + V_{J_i}\left(y_i(k), u_i(k)\right)$$

Objective Function

$$\begin{cases} -\delta\sigma_{\Delta s_i} < s_i\left(k+N_p\right) - s_{d,i}\left(k+N_p\right) < \delta\sigma_{\Delta s_i} \\ -\delta\sigma_{\Delta v_i} < v_i\left(k+N_p\right) - v_{d,i}\left(k+N_p\right) < \delta\sigma_{\Delta v_i} \\ -\delta\sigma_{\Delta u_i} < T_i\left(k+N_p\right) - T_{d,i}\left(k+N_p\right) < \delta\sigma_{\Delta u_i} \end{cases}$$

Probabilistic Constraints

Taguchi's Robustness: Insensitive to Variation

$$E_{\mathcal{L}(Y)} = \sum_{i=1}^{m} w_i \left[\left(\mu_{Y_i} - T_{ri}\right)^2 + \sigma_{Y_i}^2\right]$$

Uncertainty Quantification: First Order Second Moment

$$Y = g(Z) \approx g\left(\mu_Z\right) + \nabla g\left(\mu_Z\right) \cdot \left(Z - \mu_Z\right)$$

$$\sigma_Y^2 = \sum_{l=1}^{n} \left(\frac{\partial g}{\partial Z_l}\right)^2 \Bigg|_{\mu_z} \sigma_{Z_l}^2$$

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Simulation Results (1)

Two Simulation Settings

- The whole simulation time is 48 s with the simulation time step of 0.2 s;
- The leading vehicle starts with an initial speed of 25 m/s, and it decelerates from 3 to 5 s with -4 m/s^2 and accelerate from 27 to 35 s with 1 m/s^2;
- The initial velocity of all followers is 30 m/s, and the initial spacing error between any adjacent vehicles are 0 m;
- The maximum and minimum acceleration for all followers are 1.5 m/s^2 and -8 m/s^2, respectively.

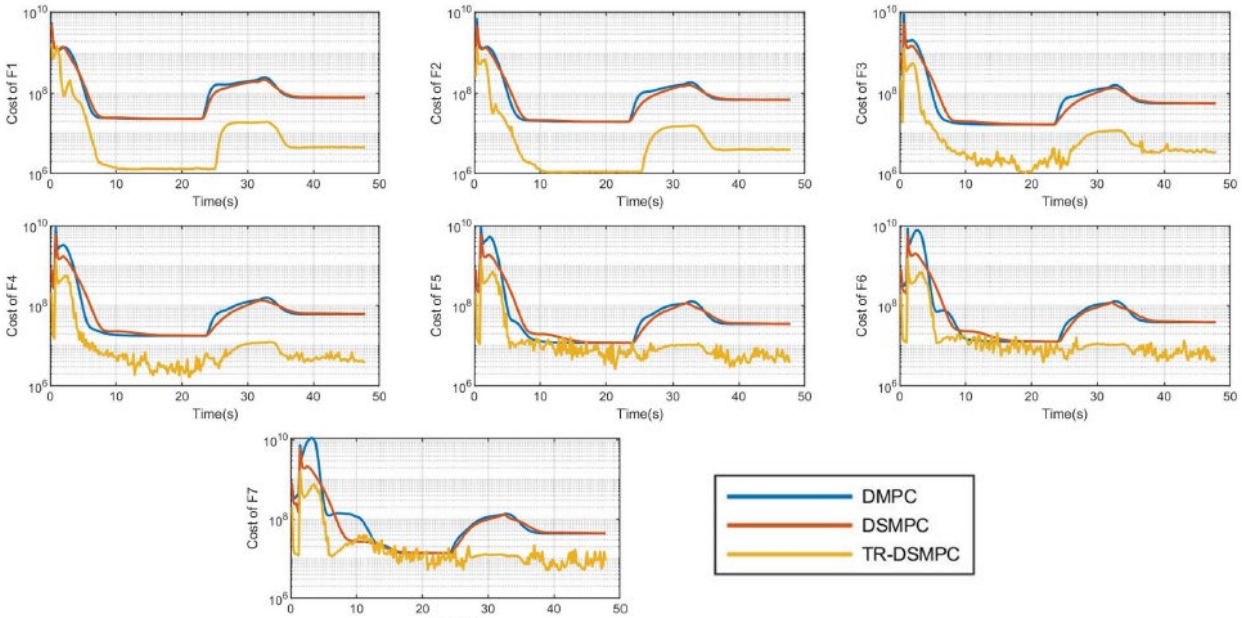
Information on Parameters and Random Variables

Random Variables	Distribution	Mean	Standard deviation
$m_{1,\ldots,7}$ (kg)	Deterministic	$(2.5, 2.3, 2.0, 1.9, 1.7, 1.6, 1.5)\times 10^3$	-
$R_{1,\ldots,7}$ (m)	Deterministic	(0.45, 0.45, 0.3, 0.3, 0.27, 0.24, 0.23)	-
$\eta_{1,\ldots,7}$	Uniform	0.92	0.0346
$C_{A,1,\ldots,7}(\mathrm{N}\cdot\mathrm{s}^2\mathrm{m}^{-2})$	Normal	(0.9, 0.9, 0.7, 0.7, 0.45, 0.45, 0.45)	(0.135, 0.135, 0.105, 0.105, 0.0675, 0.0675, 0.0675)
$\tau_{1,\ldots,7}$(s)	Uniform	0.5	0.1732
$f_{1,\ldots,7}$	Normal	0.01	0.001

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Simulation Results (2)

- **Scenario 1: Time-independent randomness under PF Topology**
  - **Control effects of relative velocity and spacing error**



MEAN COST AT TWO TIME PERIODS FOR SIMULATION 1

Follower	Mean cost ($\times 10^7$)					
	15–27 s			38–48 s		
	DMPC	DSMPC	TR-DSMPC	DMPC	DSMPC	TR-DSMPC
1	5.547	4.421	**0.251**	7.911	7.939	**0.450**
2	3.816	2.915	**0.155**	6.926	6.953	**0.396**
3	2.960	2.254	**0.213**	5.676	5.738	**0.369**
4	2.783	2.277	**0.345**	6.210	6.330	**0.484**
5	1.906	1.589	**0.699**	3.540	3.616	**0.530**
6	1.830	1.651	**0.887**	3.904	4.0	**0.612**
7	1.876	1.743	**1.130**	4.381	4.513	**0.858**

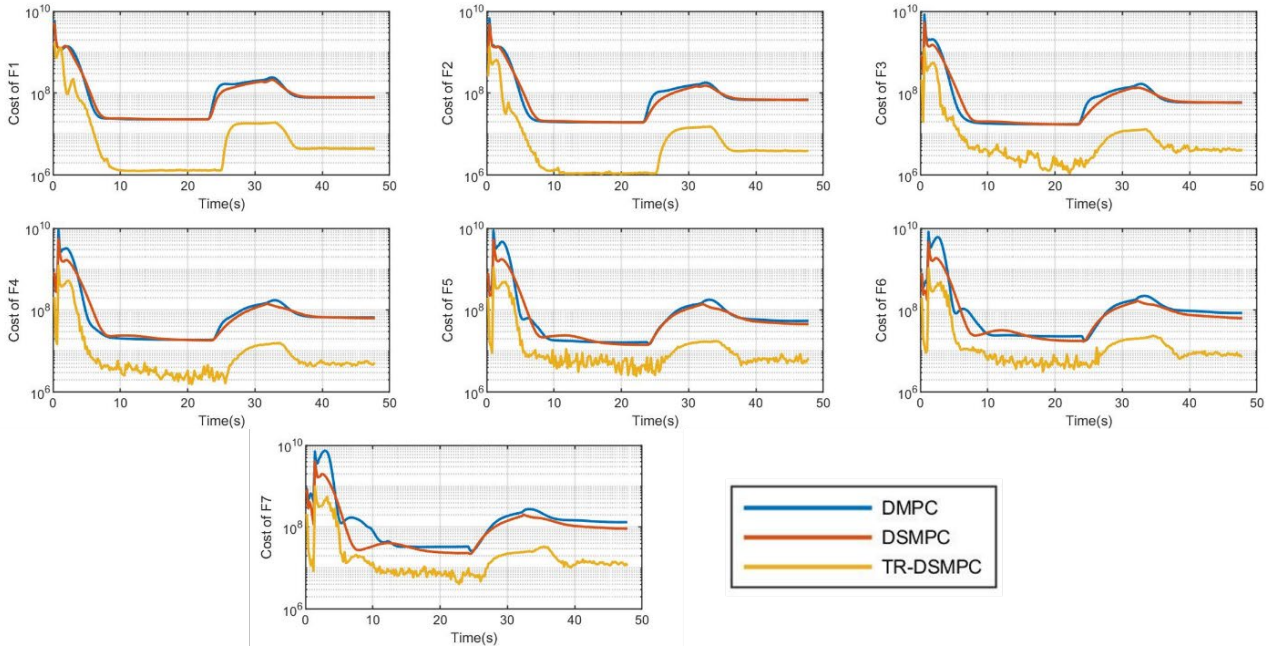
MAXIMUM AND AVERAGE OF STATISTICAL PERFORMANCE FOR SIMULATION 1

Follower	Methods	Spacing Error (m)				Relative velocity (m/s)			
		Mean		Std		Mean		Std	
		Average	Max	Average	Max	Average	Max	Average	Max
1	DMPC	0.3169	5.1453	0.1308	0.4150	0.3144	5.0	0.0407	0.3439
	DSMPC	0.2079	3.9658	0.0974	0.5767	0.2583	5.0	0.0477	0.7088
	TR-DSMPC	0.2377	**3.9751**	**0.0752**	0.5569	**0.2647**	5.0	**0.0453**	0.6772
2	DMPC	0.0939	0.7094	0.1565	0.4906	0.0401	0.6917	0.0510	0.4443
	DSMPC	0.1029	1.8674	0.1044	0.6129	0.0956	1.6206	0.0546	0.7764
	TR-DSMPC	0.1268	1.8731	**0.0944**	0.5892	0.0908	1.6499	0.0571	0.7295
3	DMPC	0.0923	0.9107	0.1692	0.6793	0.0460	0.9119	0.0618	0.6169
	DSMPC	0.1237	1.7087	0.1039	0.6524	0.0818	1.2571	0.0594	0.7826
	TR-DSMPC	0.1408	1.7355	**0.1036**	**0.6214**	0.0845	1.2375	0.0676	0.7483
4	DMPC	0.1021	1.2009	0.1671	0.6630	0.0560	1.0958	0.0626	0.6491
	DSMPC	0.1450	1.6670	0.1209	0.6321	0.0843	1.2023	0.0742	0.7518
	TR-DSMPC	0.1485	1.6258	**0.1076**	**0.5475**	0.0841	1.1727	0.0834	0.6919
5	DMPC	0.1285	1.7713	0.1774	0.9524	0.0774	1.3873	0.0801	0.9038
	DSMPC	0.1734	1.9551	0.1311	0.9066	0.1010	1.2388	0.0962	0.9177
	TR-DSMPC	0.1946	2.0354	0.1695	**0.7766**	0.1133	1.2833	0.1344	**0.8586**
6	DMPC	0.1379	1.9507	0.2111	1.2898	0.0919	1.4281	0.120	1.1396
	DSMPC	0.2158	2.0790	0.2004	1.0535	0.1242	1.1925	0.1353	0.9804
	TR-DSMPC	0.2239	2.0034	0.2303	**1.0119**	0.1327	1.1884	0.1788	**0.9050**
7	DMPC	0.3028	2.4295	0.5128	2.8870	0.1685	1.5240	0.2452	1.4290
	DSMPC	0.3057	2.1530	0.3575	2.0190	0.1641	1.2230	0.2047	1.0563
	TR-DSMPC	0.3472	**2.1306**	0.4639	2.4693	0.1786	1.2504	0.2723	**0.9294**

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Simulation Results (3)

- **Scenario 2: Time-dependent randomness under PF Topology**
  - **Control effects of relative velocity, and spacing error**



MEAN COST AT TWO TIME PERIODS FOR SIMULATION 2

Follower	Mean cost ($\times 10^7$)					
	15–27 s			38–48 s		
	DMPC	DSMPC	TR-DSMPC	DMPC	DSMPC	TR-DSMPC
1	5.570	4.458	**0.253**	7.902	7.925	**0.448**
2	3.864	2.954	**0.158**	6.920	6.952	**0.393**
3	2.954	2.340	**0.228**	5.973	6.018	**0.421**
4	2.690	2.393	**0.278**	6.709	6.668	**0.498**
5	2.068	1.971	**0.504**	5.720	5.033	**0.610**
6	2.539	2.351	**0.536**	9.167	7.225	**0.839**
7	3.490	2.927	**0.725**	14.045	10.175	**1.316**

MAXIMUM AND AVERAGE OF STATISTICAL PERFORMANCE FOR SIMULATION 2

Follower	Methods	Spacing Error (m)				Relative velocity (m/s)			
		Mean		Std		Mean		Std	
		Average	Max	Average	Max	Average	Max	Average	Max
1	DMPC	0.3109	5.1444	0.1252	0.4028	0.3165	5.0002	0.0474	0.3435
	DSMPC	0.210	3.9979	0.0948	0.5576	0.2597	5.0002	0.0545	0.7024
	TR-DSMPC	0.2330	**4.0115**	**0.0759**	0.5629	**0.2669**	5.0002	**0.0507**	0.7227
2	DMPC	0.0927	0.5862	0.1434	0.5330	0.0364	0.6570	0.0565	0.4739
	DSMPC	0.0966	1.6836	0.0959	0.6303	0.0879	1.4395	0.0590	0.7928
	TR-DSMPC	0.1088	1.6778	**0.0871**	0.6002	0.0814	1.4684	**0.0580**	0.7651
3	DMPC	0.1449	0.8857	0.1652	0.6127	0.0508	0.8939	0.0645	0.5654
	DSMPC	0.1686	1.6943	0.1057	0.6747	0.0806	1.3382	0.0672	0.7490
	TR-DSMPC	0.1946	1.7005	**0.1047**	**0.5931**	0.0848	1.2920	0.0690	0.6890
4	DMPC	0.1845	1.2599	0.1752	0.6308	0.0653	1.1295	0.0684	0.5564
	DSMPC	0.2249	1.6680	0.1059	0.6346	0.0824	1.1845	0.0716	0.6291
	TR-DSMPC	**0.2189**	1.6307	0.1134	**0.5631**	0.0850	**1.1310**	0.0838	**0.5931**
5	DMPC	0.3820	1.4002	0.2082	0.8695	0.0779	1.1974	0.0906	0.8018
	DSMPC	0.2959	1.4944	0.1402	0.7705	0.0854	1.0812	0.0969	0.7727
	TR-DSMPC	**0.3430**	1.5849	**0.1861**	**0.6276**	0.0980	1.0924	0.1234	**0.6863**
6	DMPC	0.4701	1.4258	0.2546	1.2467	0.0790	1.0971	0.1152	1.0965
	DSMPC	0.3642	1.3454	0.2458	0.9578	0.0897	0.9172	0.1394	0.8675
	TR-DSMPC	**0.3742**	**1.3320**	**0.2281**	**0.7248**	0.0941	0.9164	0.1503	**0.6850**
7	DMPC	0.6082	1.5853	0.4179	1.5976	0.1107	1.1159	0.1865	1.2410
	DSMPC	0.4281	1.2951	0.2773	1.0506	0.1011	0.8784	0.1604	0.8722
	TR-DSMPC	**0.4129**	**1.3132**	**0.2678**	**0.8514**	0.1082	0.9699	0.1708	0.7499

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Conclusions and Future Work

- **Main Results**
  - Proposed a new control methodology (TR-DSMPC) for longitudinal heterogeneous vehicle platooning by introducing Taguchi's robustness into distributed stochastic model predictive control
  - The first order second moment (FOSM) was used for uncertainty quantification, which is more efficient than Monte Carlo-based methods
  - Demonstrated that TR-DSMPC can successfully reduce the performance variations by maintaining the mean performance
  - The cost comparison between different methods showed that the robustness is improved by minimizing the mean performance and its variation
- **Future Work**
  - Study the influence of different information flow topologies on the control performance
  - Investigate the performance of TR-DSMPC for more complex application scenarios (e.g., platooning with both longitudinal and lateral motions)
  - The uncertainty from vehicle communications and the dependency between state variables can be considered and incorporated into the proposed approach

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Thanks!

Any Suggestions/Comments are welcome!

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

LingXi Li (distributed stochastic mode predictive control)

- Accounting for stochasticity (minimizing delay while reducing stochasticity effect)
- Using Taguchi's robustness
- Uncertainty modeling
  - Time-dependent modeling
  - Karhunen-Loeve (K-L) expansion

Discussion

- Hesham: why do you need the torque
  - Assuming uncertainty comes from the torque
- Simona: How to solve the optimization
  - The time step shown is for the simulation.
- Chumin: have you looked at energy efficiency?
  - No. that would be a bit more complicated
- Chumin: RL outperformed model-based control
  - RL to data, and data to RL
- Weibin: using low performance as a base rather than the average, and platoon based on characteristics (trucks in front, not back)
  - Showed how to accommodate that in the formulation
  - The average he taked about was the average of the platoon, not the individual vehicles (which would have different control each)

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.

Lingxi Li received his Ph.D. degree in electrical and computer engineering from the University of Illinois at Urbana-Champaign in 2008. Since August 2008, he has been with Indiana University-Purdue University Indianapolis (IUPUI), where he is currently a Professor of Electrical and Computer Engineering. Dr. Li's research interests include modeling, control, and optimization of complex systems, intelligent transportation systems, connected and automated vehicles, human-machine interaction, and parallel intelligence. He has authored/co-authored over one book and 130+ research articles in refereed journals and conferences. Dr. Li received a number of awards including five best paper awards, the 2021 ITSS outstanding application award, the 2017 outstanding research contributions award, the 2012 outstanding editorial service award to T-ITS, and several university research/teaching awards. He is currently serving as an associate editor for five international journals and has served as the General Chair, Program Chair, Program Co-Chair, Publication Chair, etc., for 20+ international conferences. Dr. Li is a senior member of the IEEE and a member of the SAE and AAAS.

Authorized licensed use limited to: BEIJING INSTITUTE OF TECHNOLOGY. Downloaded on March 28,2026 at 05:57:41 UTC from IEEE Xplore. Restrictions apply.